Stochastic Calculus and Option Pricing
Lecture 13 Exercises

13 Week 13 — Pricing, Hedging, and Model Risk

Focus. This variation uses linear algebra to test what a hedge can span when the model has more than one independent source of risk.

Exercise 13.1 (Count shocks before claiming a perfect hedge)**.** At a given instant, a model has d independent Brownian risk factors. There are m traded hedging instruments, each with a vector of exposures to those factors.

(a) Express the local hedging problem as a linear system.

(b) Give the rank condition for being able to cancel every local risk direction.

(c) If the rank is smaller than d and the target has exposure outside the span of the instruments, explain why fitting today's derivative price cannot remove that unspanned risk.